

IFRS 9 ECL Provisioning — Design Choices

(1-page)

Part 3 (optional extension) of the Credit Decision Data Platform. A dimensional risk-data model

that computes Expected Credit Loss (**ECL = PD × LGD × EAD**, discounted, scenario-weighted) with 3-stage classification, point-in-time vs through-the-cycle parameter versioning, and a full audit trail. It is **runnable** — `run_demo.py` executes the schema, seed, and every sample query in DuckDB.

Dimensional model & grain. Slow attributes live in dimensions (`dim_exposure`, `dim_rating_grade`, `dim_stage`, `dim_scenario`); time-varying facts are captured at a **monthly grain** (`fact_exposure_snapshot` = one row per exposure per reporting date). Everything downstream keys off (`reporting_date`, `exposure_id`), and each exposure carries the Part 1 `master_customer_id`, so the provisioning model plugs straight onto the golden record from the decision platform.

PiT vs TTC parameter versioning (the centrepiece). Risk parameters are **immutable**, effective-dated versions in `risk_parameter_set`, **split by** `parameter_basis`: **PiT** sets are macro-conditioned (tied to a `dim_scenario` and reporting period — used for the forward-looking IFRS 9 ECL) while **TTC** sets are long-run averages (scenario-independent — used for Basel/benchmarking and stable SICR thresholds). A set is bitemporal: `effective_from/to_date` is business-time validity; `created_timestamp` is system time. The PD term structure (`pd_term_structure`, cumulative PD by grade × horizon), LGD (`lgd_parameter`, downturn) and EAD CCFs (`ccf_parameter`) all hang off a `parameter_set_id`, so switching a scenario or re-versioning a model is a new set, never an in-place edit — and every provision records exactly which versions produced it.

Staging logic (IFRS 9 5.5). `fact_staging_assessment` records the stage and why:

Stage 3 on the default definition (90-DPD backstop **or** unlikely-to-pay); Stage 2 on SICR — relative lifetime-PD deterioration ≥ 2.0×, 30-DPD backstop, forbearance, or watchlist; Stage 1 otherwise. The trigger is stored (`sicr_trigger`) so an examiner sees the driver, not just the outcome (`queries/01_stage_classification.sql`).

ECL computation. $EAD = drawn + CCF \times undrawn$; PD is 12-month (Stage 1) or lifetime (Stage 2), and a defaulted exposure carries **PD = 100%** (Stage 3); LGD is the downturn rate for the collateral segment; losses are discounted at the EIR to the expected time of loss. Each of the base/upside/downside scenarios is computed against **its own PiT set** and then **probability-weighted** (`scenario_weight`) into the reported provision (`queries/02_provision_calculation.sql`). On the sample book this yields the expected staging gradient — Stage 1 ≈ 0.5%, Stage 2 ≈ 7.7%, Stage 3 ≈ 48% coverage.

Audit trail & reproducibility. `audit_stage_transition` logs every period-over-period migration (direction, trigger, DPD/PD at transition) — the reconciliation regulators require.

audit_provision_calculation freezes, per provision, the model version, the exact PiT parameter_set_ids, the scenario weights, the inputs, and a **SHA-256 of those inputs** — the same tamper-evident, fully-reproducible philosophy as Part 1's immutable decision snapshots, so any provision can be re-derived and defended years later.

Deliberate simplifications (demo scope). Lifetime ECL uses a cumulative-PD-to-horizon shortcut rather than a full period-by-period marginal-PD survival curve; discounting uses the expected-time-of-loss midpoint rather than per-period discounting; LGD is held constant across scenarios (PD carries the macro sensitivity). All three are the first things to deepen for production, and none change the schema — they are richer parameter tables and a marginal-loss calculation over the same grain.